

Intrusion Prevention System (IPS) ¹ ²

An **Intrusion Prevention System (IPS)** is a network security application designed to monitor network or system activities for malicious activity and automatically block them. It evolved from Intrusion Detection Systems (IDS), which detect and report threats to the security team. An IPS not only detects and reports threats but also has automated threat prevention capabilities ¹ ².

How IPS Works

An IPS works by analyzing network traffic in real-time and comparing it against known attack patterns and signatures. When the system detects suspicious traffic, it blocks it from entering the network ². IPS solutions use three primary threat detection methods:

1. **Signature-based detection:** Analyzes network packets for attack signatures—unique characteristics or behaviors associated with a specific threat. It maintains a database of attack signatures and compares network packets against it ¹.
2. **Anomaly-based detection:** Uses artificial intelligence and machine learning to create a baseline model of normal network activity. It compares ongoing network activity to the model and responds to deviations ¹.
3. **Policy-based detection:** Based on security policies set by the security team. It blocks actions that violate these policies ¹.

There are **four major types of intrusion prevention systems**:

- Network-based intrusion prevention system (NIPS): It monitors traffic and scans for cyberthreats at key network locations.
- Wireless intrusion prevention system (WIPS): It monitors wireless networks for suspicious traffic by analyzing wireless networking protocols.
- Host-based intrusion prevention system (HIPS): It monitors a single host for suspicious activity.
- Network behavior analysis (NBA): It monitors the entire network for suspicious traffic by analyzing protocol activity.

There are also two main types of intrusion detection systems: Network-based Intrusion Detection Systems (NIDS) and Host-based Intrusion Detection Systems (HIDS).

What Is An Intrusion Detection System (IDS)?

An intrusion detection system (IDS) is an application that monitors network traffic and searches for known threats and suspicious or malicious activity. The IDS sends alerts to IT and security teams when it detects any security risks and threats.

Most IDS solutions simply monitor and report suspicious activity and traffic when they detect an anomaly. However, some can go a step further by taking action when it detects anomalous activity, such as blocking malicious or suspicious traffic.

IDS tools typically are software applications that run on organizations' hardware or as a network security solution. There are also cloud-based IDS solutions that protect organizations' data, resources, and systems in their cloud deployments and environments.

Types Of Intrusion Detection Systems (IDS)

IDS solutions come in a range of different types and varying capabilities. Common types of intrusion detection systems (IDS) include:

1. **Network intrusion detection system (NIDS):** A NIDS solution is deployed at strategic points within an organization's network to monitor incoming and outgoing traffic. This IDS approach monitors and detects malicious and suspicious traffic coming to and going from all devices connected to the network.
2. **Host intrusion detection system (HIDS):** A HIDS system is installed on individual devices that are connected to the internet and an organization's internal network. This solution can detect packets that come from inside the business and additional malicious traffic that a NIDS solution cannot. It can also discover malicious threats coming from the host, such as a host being infected with malware attempting to spread it across the organization's system.
3. **Signature-based intrusion detection system (SIDS):** A SIDS solution monitors all packets on an organization's network and compares them with attack signatures on a database of known threats.

1. **Monitoring the performance** of key firewalls, files, routers, and servers to detect, prevent, and recover from cyberattacks
2. **Enabling system administrators** to organize and understand their relevant operating system audit trails and logs that are often difficult to manage and track
3. **Providing an easy-to-use interface** that allows staff who are not security experts to help with the management of an organization's systems
4. **Providing an extensive database** of attack signatures that can be used to match and detect known threats
5. **Providing a quick and effective reporting system** when anomalous or malicious activity occurs, which enables the threat to be passed up the stack
6. **Generating alarms** that notify the necessary individuals, such as system administrators and security teams, when a breach occurs
7. **In some cases, reacting to potentially malicious actors** by blocking them and their access to the server or network to prevent them from carrying out any further action

How Does an IPS Work?

An IPS works by analyzing network traffic in real-time and comparing it against known attack patterns and signatures. When the system detects suspicious traffic, it blocks it from entering the network.

Types of IPS

There are two main types of IPS:

1. **Network-Based IPS:** A Network-Based IPS is installed at the network perimeter and monitors all traffic that enters and exits the network.
2. **Host-Based IPS:** A Host-Based IPS is installed on individual hosts and monitors the traffic that goes in and out of that host.

1. Network-based intrusion prevention system (NIPS):

It monitors the entire network for suspicious traffic by analyzing protocol activity.

2. Wireless intrusion prevention system (WIPS):

It monitors a wireless network for suspicious traffic by analyzing wireless networking protocols.

3. Network behavior analysis (NBA):

It examines network traffic to identify threats that generate unusual traffic flows, such as distributed denial of service attacks, specific forms of malware and policy violations.

4. Host-based intrusion prevention system (HIPS):

It is an inbuilt software package which operates a single host for doubtful activity by scanning events that occur within that host.

- **Network Intrusion Detection System (NIDS):** Network intrusion detection systems (NIDS) are set up at a planned point within the network to examine traffic from all devices on the network. It performs an observation of passing traffic on the entire subnet and matches the traffic that is passed on the subnets to the collection of known attacks. Once an attack is identified or abnormal behavior is observed, the alert can be sent to the administrator. An example of a NIDS is installing it on the subnet where [firewalls](#) are located in order to see if someone is trying to crack the [firewall](#).
- **Host Intrusion Detection System (HIDS):** Host intrusion detection systems (HIDS) run on independent hosts or devices on the network. A HIDS monitors the incoming and outgoing packets from the device only and will alert the administrator if suspicious or malicious activity is detected. It takes a snapshot of existing system files and compares it with the previous snapshot. If the analytical system files were edited or deleted, an alert is sent to the administrator to investigate. An example of HIDS usage can be seen on mission-critical machines, which are not expected to change their layout.

VoIP PBX security

- To ensure **VoIP PBX security**, consider the following best practices:
- **Identify Common Vulnerabilities:** Understand the most common VoIP security issues, such as eavesdropping, denial of service attacks, and unauthorized access.
- **implement Security Policies:** Establish robust security policies and practices, including user authentication and access controls.
- **Choose a Secure Provider:** Select a VoIP provider that complies with security protocols and offers encryption for data transmission.
- **Utilize Advanced Settings:** Use advanced PBX settings like SRTP (Secure Real-time Transport Protocol) and outbound call restrictions to enhance security.

Components of VoIP (Voice over Internet Protocol) include:

- Session Border Controller (SBC)
- Media Server
- Application Server
- Database Services
- SIP services
- IP PBX
- Endpoint devices (phone)
- IP network
- End-user equipment
- Network components
- Call processors
- Gateways and protocols
- VoIP routers
- Power over Ethernet (PoE) switches
- Session border controllers (SBC)
- Uninterruptible power supply (UPS)

Common vulnerabilities in VoIP (Voice over Internet Protocol) systems include: 

1. **Denial of Service (DoS) Attacks:** Overwhelming VoIP servers or networks with traffic, rendering them unavailable.
2. **Malware and Viruses:** Threats that can compromise VoIP systems.
3. **Eavesdropping or Call Interception:** Unauthorized access to call content.
4. **Toll Fraud:** Unauthorized use of VoIP services.
5. **Phishing (Vishing):** Social engineering attacks targeting VoIP users.
6. **Spam over Internet Telephony (SPIT):** Unsolicited messages.
7. **Misconfigured Internet Telephones (VoMIT):** Configuration errors leading to vulnerabilities.

Countermeasures:

- Choose a trusted VoIP provider with a good track record.
- Admin access should be implemented in a controlled and careful manner.
- VPNs should be used in case of remote access.
- Detailed network tests should be done periodically.
- VPNs should be used and their endpoint filtering should be enabled.
- Regularly checking history, access, and call logs.
- Passwords used should be strong.
- 2FA should be enabled.
- VoIP firmware and OS should always be up to date.
- Organizations should have more security professionals.

What is a private branch exchange (PBX)?

A private branch exchange (PBX) is a telephone system within an enterprise that switches calls between users on local lines, while enabling all users to share a certain number of external phone lines. In contrast to a public switched telephone network, the main purpose of a PBX is to save the cost of requiring a line for each user to the telephone company's central office.

Used as a business telephone system or private telephone network, a PBX is owned and operated by the enterprise rather than the telephone company. However, the telephone company may be considered a supplier or service provider. Originally, private branch exchanges used analog technology. Today, PBXs use digital technology -- digital signals are converted to analog for outside calls on the local loop using [Plain Old Telephone Service](#). Nonetheless, PBXs can include network switching systems that accommodate analog phones into the enterprise's digital PBX system.

What is Telecom Expense Management (TEM)?

Telecom Expense Management (or Technology Expense Management) is the management of complex Enterprise telecom-related expenses and to do this right is an ongoing and proactive process. The goal is to embrace change and understand that constant effort to improve is necessary to avoid complacency.

This practice of fully leveraging the latest and greatest in technology to improve all areas of business is commonly referred to as **digital transformation** and it plays a very central role in current enterprise expense management best practices.

This management focus helps large enterprises improve their ability to manage their accounts payable (AP) process, the monitoring and reporting of billing charges, contracts, network changes, and service inventory to name a few areas.

Security in Operating Systems ¹ ² ³

Operating system security is crucial for protecting computer resources such as CPU, memory, disk, software programs, and data from unauthorized access and malicious activities. It involves implementing measures to ensure the availability, confidentiality, and integrity of the operating system.

Key Principles of OS Security

Integrity: Ensures that unauthorized users cannot alter the system's vital files and resources. Users with insufficient privileges should not be able to modify critical system files ¹.

Secrecy: Only authorized users should have access to the system's objects. System files should not be accessible to everyone ².

Availability: All authorized users must have access to system resources. No single user or process should monopolize resources, leading to service denial ¹.

Common Threats to OS Security

Malware: Malicious software designed to harm computer systems, networks, or users. It includes viruses, worms, Trojan horses, and other dangerous software ¹.

Network Intrusion: Unauthorized access to a system by masqueraders, misfeasors, or rogue users. Intrusion detection systems (IDS) monitor network traffic for malicious activities ².

Buffer Overflow: A technique where an intruder exploits a vulnerability in a program to execute malicious code. This can lead to security breaches in mail servers and other web servers ¹.

Classic Security Models

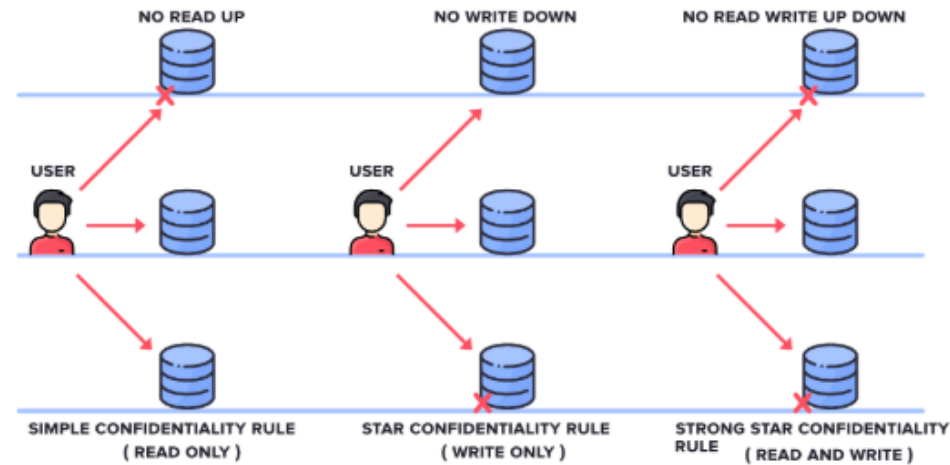
These models are used for maintaining goals of security, i.e. Confidentiality, Integrity, and Availability. In simple words, it deals with **CIA Triad** maintenance. There are 3 main types of Classic Security Models.

- **Bell-LaPadula**
- **Biba**
- **Clarke Wilson Security Model**

1. Bell-LaPadula

This Model was invented by Scientists **David Elliot Bell** and **Leonard J. LaPadula**. Thus this model is called the Bell-LaPadula Model. This is used to maintain the **Confidentiality** of Security. Here, the classification of Subjects(Users) and Objects(Files) are organized in a non-discretionary fashion, with respect to different layers of secrecy.

BELL - LAPADULA MODEL

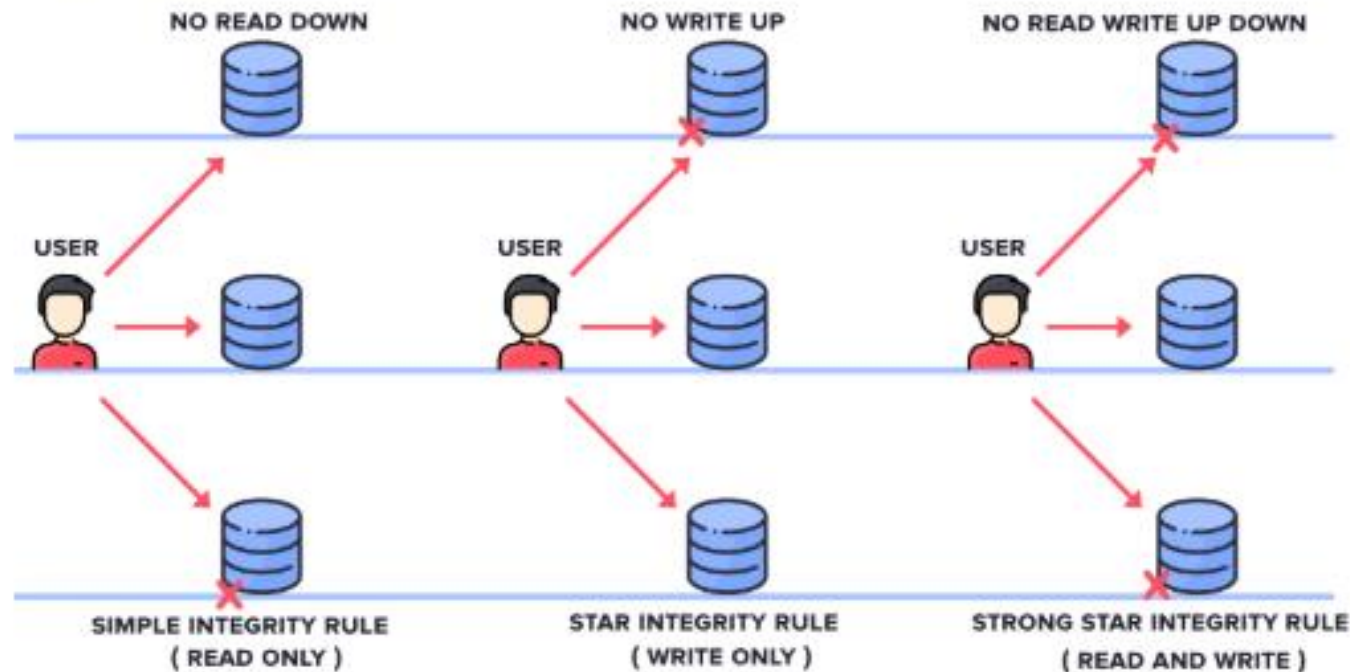


- **SIMPLE CONFIDENTIALITY RULE:** Simple Confidentiality Rule states that the Subject can only **Read** the files on the Same Layer of Secrecy and the Lower Layer of Secrecy but not the Upper Layer of Secrecy, due to which we call this rule as **NO READ-UP**
- **STAR CONFIDENTIALITY RULE:** Star Confidentiality Rule states that the Subject can only **Write** the files on the Same Layer of Secrecy and the Upper Layer of Secrecy but not the Lower Layer of Secrecy, due to which we call this rule as **NO WRITE-DOWN**
- **STRONG STAR CONFIDENTIALITY RULE:** Strong Star Confidentiality Rule is highly secured and strongest which states that the Subject can **Read** and **Write** the files on the Same Layer of Secrecy only and not the Upper Layer of Secrecy or the Lower Layer of Secrecy, due to which we call this rule as **NO READ WRITE UP DOWN**

2. Biba

This Model was invented by Scientist **Kenneth .J. Biba**. Thus this model is called Biba Model. This is used to maintain the **Integrity** of Security. Here, the classification of Subjects(Users) and Objects(Files) are organized in a non-discretionary fashion, with respect to different layers of secrecy. This works the exact reverse of the Bell-LaPadula Model.

BIBA MODEL

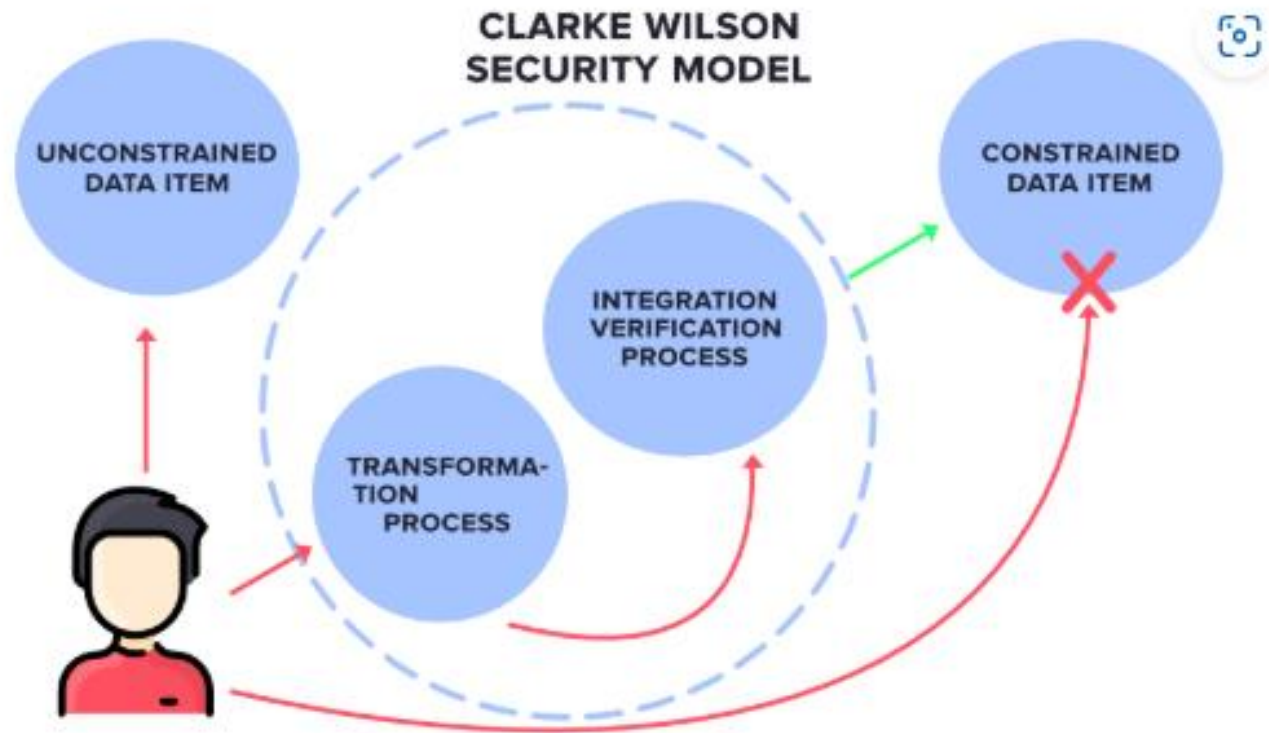


It has mainly 3 Rules:

- **SIMPLE INTEGRITY RULE:** Simple Integrity Rule states that the Subject can only **Read** the files on the Same Layer of Secrecy and the Upper Layer of Secrecy but not the Lower Layer of Secrecy, due to which we call this rule as **NO READ DOWN**
- **STAR INTEGRITY RULE:** Star Integrity Rule states that the Subject can only **Write** the files on the Same Layer of Secrecy and the Lower Layer of Secrecy but not the Upper Layer of Secrecy, due to which we call this rule as **NO WRITE-UP**
- **STRONG STAR INTEGRITY RULE**

3. Clarke Wilson Security Model

This Model is a highly secured model. It has the following entities.



- **SUBJECT:** It is any user who is requesting for Data Items.
- **CONSTRAINED DATA ITEMS:** It cannot be accessed directly by the Subject. These need to be accessed via **Clarke Wilson Security Model**
- **UNCONSTRAINED DATA ITEMS:** It can be accessed directly by the Subject.

Security for cyber systems built without a trustworthy Operating System (OS) is simply a scientific impossibility. The proven scientific principle of the “Reference Monitor”, whose implementation is called a “security kernel”, enables engineering a verifiably secure OS. The Reference Monitor [3] is an abstraction that allows active entities called subjects to make reference to passive entities called objects, based on a set of current access authorizations. A security kernel (hardware and software) ensures that every reference by a subject to information in an object (e.g., by a processor to primary memory) or change of authorization must go through the Reference Monitor. This mediation by the Reference Monitor is depicted in Figure 1.

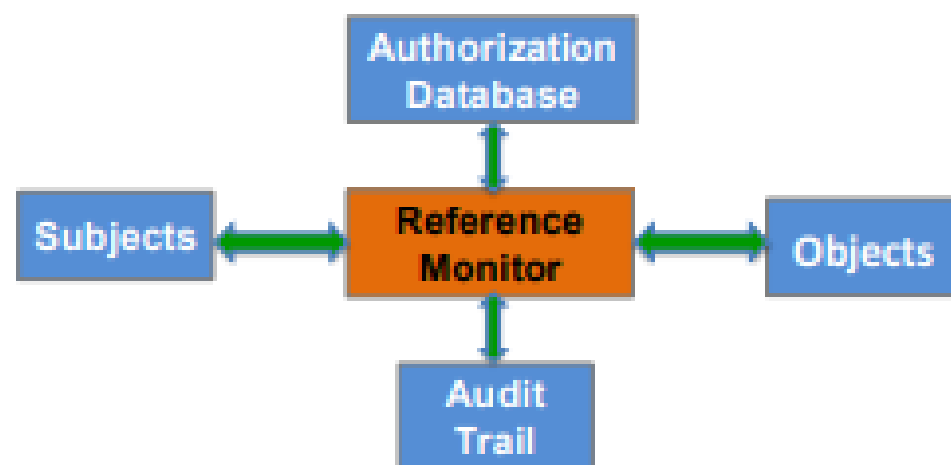


Figure 1. Reference Monitor concept.

- Trust models are frameworks that define how entities in a network can verify each other's identities and credentials. They are essential for network security, as they enable secure communication and data exchange among parties that may not have direct or prior relationships. In this article, you will learn about the different types of trust models and how they work in network security, with a focus on SSL certificates and trust models.

1 Direct Trust

Direct trust is the simplest and most basic form of trust model, where each entity in a network trusts only itself and those that it has explicitly verified. For example, if Alice wants to communicate with Bob, she needs to have a copy of Bob's public key and verify that it belongs to him. This way, Alice can encrypt and decrypt messages with Bob using his public key and her private key, and vice versa. Direct trust is suitable for small and closed networks, where the number of entities is limited and the trust relationships are stable.

2 Third-Party Trust

Third-party trust is a more common and scalable form of trust model, where each entity in a network trusts a central authority that vouches for the identities and credentials of other entities. For example, if Alice wants to communicate with Bob, she does not need to have a copy of Bob's public key or verify it herself. Instead, she can rely on a trusted third party, such as a certificate authority (CA), that issues and signs digital certificates for Bob and other entities. A digital certificate is a document that contains the public key and other information of an entity, such as its name, domain, and expiration date. A CA is a trusted entity that verifies the identity and legitimacy of

an entity before issuing and signing a digital certificate for it. This way, Alice can verify Bob's identity and public key by checking his digital certificate and the signature of the CA. Third-party trust is suitable for large and open networks, where the number of entities is dynamic and the trust relationships are diverse.

3 Hierarchical Trust

Hierarchical trust is a variation of third-party trust, where there is a hierarchy of CAs that delegate their trust to lower-level CAs. For example, if Alice wants to communicate with Bob, she does not need to trust the CA that issued and signed Bob's digital certificate directly. Instead, she can trust a higher-level CA that has issued and signed the certificate of the CA that issued and signed Bob's certificate. This way, Alice can verify Bob's identity and public key by following a chain of certificates and signatures from the highest-level CA to the lowest-level CA. Hierarchical trust is suitable for complex and distributed networks, where the CAs are organized into different domains, regions, or levels.

4 Web of Trust

Web of Trust is another variation of third-party trust, where there is no central authority or hierarchy of CAs, but rather a network of peers that endorse and rate each other's identities and credentials. For example, if Alice wants to communicate with Bob, she does not need to trust a CA or a chain of CAs. Instead, she can trust other peers that have verified and vouched for Bob's identity and public key. These peers can be friends, colleagues, or other entities that Alice knows and trusts. Alice can also check the reputation and trustworthiness of these peers based on their ratings, feedback, and history. Web of Trust is suitable for decentralized and peer-to-peer networks, where the CAs are not available, reliable, or trustworthy.

5 Hybrid Trust

Hybrid trust is a combination of different trust models, where each entity in a network can choose the most appropriate trust model for its needs and preferences. For example, if Alice wants to communicate with Bob, she can use direct trust, third-party trust, hierarchical trust, or web of trust, depending on the situation, the availability of CAs or peers, and the level of security and convenience she requires. Hybrid trust is suitable for heterogeneous and adaptive networks, where the entities have different capabilities, requirements, and expectations.

International standards for operating systems

- **Authentication.** It ensures that only the right people and devices can access a network. This includes enforcing strong passwords and requiring users to [enable two-factor authentication](#) for an extra layer of protection.
- **Virtual private network (VPN).** A [VPN](#) secures the connection between a user's device and a network, especially when accessing it remotely. VPNs are critical for maintaining security when employees work from home or connect over public Wi-Fi.
- **Network encryption.** Network encryption (including [VPN encryption](#)) protects sensitive traffic as it moves across the network by converting it into unreadable code. This ensures that even if someone intercepts the data, it can't understand it without the decryption key.
- **Firewalls.** [Firewalls](#) act as the first line of defense between a network and potential threats. They monitor and control both incoming and outgoing traffic, blocking unauthorized access while letting legitimate traffic pass through without issues.